07 VirtualBox Networking Modes

1. What Is Virtualization?

Virtualization is a technology that allows one physical computer to run multiple virtual computers.

Instead of purchasing a separate physical computer for every operating system, we can create virtual machines (VMs) on one computer.

Example

Suppose you have a Windows laptop. Using VirtualBox, you can run:

- Kali Linux for cybersecurity practice.
- Ubuntu for Linux commands and networking.
- Windows in a separate virtual environment for testing.

All these systems can run on the same physical computer, depending on its hardware resources.

Advantages of virtualization

- Saves money and hardware resources.
- Allows multiple operating systems to run on one computer.
- Provides an isolated environment for testing.
- Makes it easy to create, restore and copy lab environments.
- Helps students practice cybersecurity safely.

What Is a Hypervisor?

A **hypervisor** is software that creates and manages virtual machines.

There are two main types:

Type	Meaning	Example
Type 1	Runs directly on the physical hardware	VMware ESXi, Microsoft Hyper-V
Type 2	Runs as an application on a host operating system	Oracle VirtualBox, VMware Workstation

Remember: Virtualization allows one physical computer to run multiple virtual systems.

2. What Is a Virtual Machine (VM)?

A **virtual machine is a software-based computer** that behaves like a physical computer.

A VM has virtual hardware, such as:

- Virtual CPU
- Virtual RAM
- Virtual hard disk
- Virtual network adapter
- Virtual display and other devices

It runs its own operating system and applications.

Example of a cybersecurity lab

Virtual machine	Purpose
Kali Linux	Security testing and cybersecurity tools
Ubuntu	Linux commands, networking and server administration
Windows 10/11 evaluation VM	Windows administration, event logs and security practice

These VMs can communicate through a virtual network when their network settings allow it.

Host vs Guest

Term	Meaning
Host machine	The physical computer running VirtualBox
Guest machine	The operating system running inside a VM

Example: If your laptop runs Windows and VirtualBox runs Ubuntu, Windows is the host OS and Ubuntu is the guest OS.

3. VirtualBox Networking Modes

VirtualBox networking modes control how a VM communicates with other VMs, the host computer and external networks.

The four important modes are:

1. NAT
2. NAT Network
3. Host-only Adapter
4. Bridged Adapter

The correct mode depends on what you want to achieve in your lab.

4. NAT (Network Address Translation)

NAT allows a virtual machine to access external networks, usually the internet, through the host computer's network connection.

VirtualBox translates the VM's network traffic so it can communicate externally.

How NAT works

1. The VM sends a request to the internet.
2. VirtualBox translates the network traffic.
3. The request goes through the host's network connection.
4. The response returns to the VM.

Example

Your Ubuntu VM needs to install software.

Run:

sudo apt update

With working internet connectivity and appropriate network configuration, the VM can contact Ubuntu's software repositories through NAT.

Advantages

- Simple to configure.
- Allows internet access.
- The VM is generally not directly reachable from other devices on the physical network.
- Useful for downloading updates and installing packages.

Limitations

- Other VMs on separate default NAT networks generally cannot communicate directly with each other.
- Inbound connections require additional configuration, such as port forwarding.
- It does not guarantee complete security or isolation.

When to use NAT

Use NAT when you want a VM to access the internet but do not need direct communication between multiple VMs.

Example: Use NAT to update Ubuntu or Kali Linux.

5. NAT Network

NAT Network allows multiple virtual machines to communicate with each other and access external networks through NAT.

Unlike separate default NAT configurations, VMs connected to the same NAT Network can generally communicate directly with one another.

How NAT Network works

Suppose you create a NAT Network named CyberLab.

Connect these VMs to it:

- Kali Linux
- Ubuntu
- Windows

They can communicate with each other over the virtual network, and they can usually access the internet through NAT.

Advantages

- Allows communication between VMs on the same NAT Network.
- Supports internet access.
- Useful for multi-machine labs.
- Keeps the VMs separated from direct attachment to the physical LAN.

Limitations

- Network services and routing must be configured correctly.
- Inbound connectivity from the host or physical LAN may require additional configuration.
- Internet access can still expose the lab to external threats.

When to use NAT Network

Use it when you want multiple VMs to communicate and need internet access.

Example: Kali scans a deliberately vulnerable test service running on Ubuntu in your own lab.

Only scan systems you own or have explicit permission to test.

6. Host-only Networking

Host-only networking creates a private virtual network between the host computer and connected virtual machines.

By default, this network does not provide direct internet access to the VMs.

How Host-only works

Suppose you create a host-only network called CyberLab-HostOnly.

Connect Kali and Ubuntu to this network.

They can communicate with each other, and the host can communicate with the VMs through the host-only interface.

Advantages

- Useful for isolated cybersecurity practice.
- Allows communication between the host and VMs.
- Allows connected VMs to communicate with each other.
- Does not normally provide direct internet access.

Limitations

- Internet access is not provided by default.
- Incorrect routing or additional adapters may reduce isolation.
- The VMs can still attack each other if vulnerable services are exposed.

When to use Host-only

Use Host-only for practicing attacks and defenses in a private environment where internet connectivity is unnecessary.

Example: Practice Linux administration, inspect logs or test a vulnerable web application between your own Kali and Ubuntu VMs.

7. Bridged Networking

Bridged networking connects a VM to the physical network through the host computer's network adapter.

The VM appears as a separate device on the physical network and may obtain an IP address from the physical network's DHCP server.

How Bridged networking works

Suppose your home router uses the network 192.168.1.0/24.

Your laptop may have the IP address 192.168.1.10, while a bridged Ubuntu VM might receive 192.168.1.20.

Both devices are connected to the same physical LAN, subject to the router and network configuration.

Advantages

- Allows a VM to behave like a separate physical device on the LAN.
- Other LAN devices may connect to services running on the VM.
- Useful for authorized testing of services on a controlled network.

Limitations

- The VM may be reachable by other devices on the physical network.
- Vulnerable services may be exposed to the home, office or institutional LAN.
- Network policies and firewall settings affect connectivity.
- Wireless bridging behavior can differ depending on the host adapter and VirtualBox configuration.

When to use Bridged networking

Use it only when the VM needs direct participation in the physical network and the exposure is understood and authorized.

Example: Testing a server VM that must be accessible to other devices on a controlled test LAN.

8. Comparison of VirtualBox Networking Modes

Feature	NAT	NAT Network	Host-only	Bridged
Internet access	Usually yes	Usually yes	No, by default	Usually yes
VM-to-VM communication	Not normally between separate default NAT networks	Yes, on the same NAT Network	Yes, on the same host-only network	Yes, if LAN rules allow
Host-to-VM communication	Inbound access usually requires port forwarding or additional configuration	Depends on configuration	Yes, through the host-only network	Usually possible, subject to network rules
Direct physical LAN exposure	Low by default	Low by default	Normally isolated from physical LAN	Potentially high
Typical use	Software updates	Multi-VM labs with internet	Isolated security labs	Controlled LAN testing
Recommended for vulnerable labs	Safer starting point for internet access	Useful with careful configuration	Usually preferred for isolated practice	Avoid unless specifically required and safely controlled

Easy way to remember

- **NAT:** One VM needs internet access.
- **NAT Network:** Multiple VMs need to communicate and access the internet.
- **Host-only:** Private communication between the host and VMs, without internet by default.
- **Bridged:** VM joins the physical network like a separate device.

9. Why Should Vulnerable VMs Never Be Bridged?

A deliberately vulnerable VM contains security weaknesses intended for learning and testing.

Examples include vulnerable web applications, outdated software and insecure configurations.

When such a VM uses bridged networking, it can become accessible to other devices on the physical LAN.

Main risks

1. **Unauthorized access:** Other devices on the network may reach vulnerable services.
2. **Malware spread:** A compromised VM may attempt to attack other systems.
3. **Data exposure:** Services may expose files, credentials or test data.
4. **Unintended scanning:** Security tools may accidentally target real devices instead of lab systems.
5. **Network compromise:** An attacker who compromises the VM may use it to attempt further attacks.

Safer approach

- Prefer a dedicated Host-only network for isolated labs.
- Use a NAT Network when VMs need to communicate and internet access is required.
- Keep vulnerable systems updated where appropriate, and isolate intentionally vulnerable targets from untrusted networks.
- Use strong host security and firewall rules.
- Disable unnecessary services and network adapters.
- Never connect deliberately vulnerable systems to public or institutional networks without explicit authorization and suitable controls.

Important: Bridged networking is not inherently malicious or always unsafe, but it increases exposure. Avoid it for intentionally vulnerable systems unless you have a carefully controlled, authorized test network.

10. What Is a Snapshot?

A **snapshot records the state of a virtual machine at a particular point in time**, allowing you to return to that state later.

It is useful before making changes that may break the VM.

Example

Before installing a tool on Ubuntu:

1. Shut down the VM if appropriate.
2. Open VirtualBox.
3. Select the Ubuntu VM.
4. Open its Snapshots view.
5. Create a snapshot named Clean-Ubuntu-Before-Testing.
6. Install and test the software.
7. Restore the snapshot if you need to return to the earlier state.

Advantages of snapshots

- Helps recover from configuration mistakes.
- Makes repeated lab exercises easier.
- Allows testing without permanently retaining every change.
- Helps restore a clean starting point.

Important limitations

- A snapshot is not a complete independent backup.
- It may depend on the original virtual disk and snapshot files.
- It can consume significant disk space.
- Restoring an old snapshot may discard subsequent changes.

For important files, keep separate backups.

11. Conclusion

Virtualization makes it possible to build a low-cost cybersecurity lab on one computer. VirtualBox networking modes determine how virtual machines communicate with each other, the host and external networks.

For beginners, Host-only networking is a good choice for isolated practice, while NAT Network is useful when multiple VMs need internet access. Bridged networking should be avoided for deliberately vulnerable systems unless the environment is specifically controlled and authorized.

Snapshots, careful network configuration and regular backups help maintain a safer and more reliable lab.